

Abdullah Murad Ertaskin

Preferred name: Murad

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Education

Adelaide University

Bachelor of Computer Science, AI/ML Major

Adelaide, Australia

Expected July 2027

- Core Subject GPA : 6.5 / 7
- Relevant coursework: Machine Learning, Evolutionary Algorithms, AI Modelling, Computer Networks, Operating Systems, Database Systems, Software Engineering, Web Development

Projects — [muradertas.dev](#)

Market Maker Simulator

[GitHub](#) | [Live Demo](#)

Python · FastAPI · React/TypeScript · WebSockets · Docker · pytest

- Implemented the Avellaneda–Stoikov (2008) stochastic control framework.
- Built a price-time priority limit order book and matching engine entirely from scratch in Python; noise-trader arrivals discretised as Bernoulli($\lambda \cdot dt$) per tick
- Two-state Markov regime-switching volatility ($\sigma_{\text{low}}/\sigma_{\text{high}}$) with hot-swappable transition probabilities; hard inventory caps with asymmetric quote suppression
- Full-stack: FastAPI async backend streaming JSON snapshots at ~ 20 Hz over WebSocket to a React / TypeScript dashboard with live Recharts panels (LOB depth, PnL, Sharpe, drawdown, adverse selection, fill heatmap); Docker Compose + Railway deployment; pytest suite covering LOB matching, A–S formula correctness, and rolling metrics

Small Language Model — Transformer vs. RNN

[GitHub](#) | [Live Demo](#)

Python · PyTorch · BPEasy

- Built Transformer and LSTM language models from scratch for sequence modelling and text generation, including Byte Pair Encoding tokenisation and custom training/evaluation loops in PyTorch. Models are trained on Shakespeare's works.
- Developed a side-by-side comparison interface to evaluate coherence, repetition, and response quality across architectures under consistent prompts and generation settings

Portfolio Website — [muradertas.dev](#)

[GitHub](#) | [muradertas.dev](#)

Next.js 14 · TypeScript · Tailwind CSS · Canvas 2D

- Built a full-stack Next.js 14 portfolio using the App Router with a responsive, accessible UI, shared design system, and dark/light mode support with seamless theme switching
- Implemented a custom Canvas WaveGrid hero (perspective projection, ripple interactions, 60 FPS optimisations) and a validated contact form API route with email delivery through Resend

Experience

Freelance Software Developer

Jul 2025 – Present

Upwork

Remote

- Delivered custom ML models, full-stack web applications, and automation scripts for clients across multiple domains

Software Engineering

Dec 2025

Forage — Commonwealth Bank Simulation

- Extended a .NET/C# backend and modified a React / Redux frontend; worked with MongoDB, Postman, TypeScript, and NPM

Software Engineering

Dec 2025

Forage — Quantum Simulation

- Built an interactive Dash application for price-elasticity analysis, enabling clients to quantify revenue impact of pricing decisions

Technical Skills

Languages: Python, JavaScript / TypeScript, Java, C++, SQL

Frameworks & Libraries: React, FastAPI, Node.js, Flask, Django, PyTorch, Pydantic, Recharts

Tools: Git, Docker, AWS, Linux, PostgreSQL, MongoDB, WebSockets, pytest, Vite

Quantitative: Stochastic processes, optimal control, Markov chains, Poisson processes, Brownian motion, game theory, optimisation

Other: CI/CD, RESTful APIs, Agile/Scrum, microservices